ICIC Express Letters
Part B: Applications
Volume **17**, Number **8(tentative)**, August **2026**

ICIC International ©2026 ISSN 2185-2766
pp. 1–SCI2025-016

A KOBERT-SVD HYBRID MODEL FOR PERSONALIZED COSMETIC RECOMMENDATIONS BASED ON USER SENTIMENT AND SKIN ATTRIBUTES

Chan Hee Kim[1], Haram Kim[2] and Dongsoo Kim[2,*]

[1]Department of Information Technology Distribution and Logistics
[2]Department of Industrial and Information Systems Engineering
Soongsil University
369 Sangdo-Ro, Dongjak-Gu, Seoul 06978, Korea
{ chan4782; haramkim68 }@soongsil.ac.kr; *Corresponding author: dskim@ssu.ac.kr

Received October 2025; accepted December 2025

Abstract. *This study presents a personalized cosmetic recommendation system that integrates online review data with sentiment analysis, skin profile filtering, and a hybrid modeling approach. A total of 11,795 refined reviews were collected and preprocessed through duplicate removal, morphological analysis, and keyword-based filtering to extract relevant skin-related attributes. Sentiment scores were generated using a rule-based method and a machine learning classifier combining TF-IDF and logistic regression, and these scores were incorporated into rating augmentation for collaborative filtering based on Singular Value Decomposition (SVD). In parallel, KoBERT embeddings were applied to textual reviews to enable context-aware content-based recommendation. Skin type, tone, and primary concerns were used as pre-filtering criteria to reduce the candidate pool and enhance personalization. The two recommendation outputs were fused through a weighted ensemble to generate final product rankings. The proposed approach demonstrates the effectiveness of combining emotional cues, textual semantics, and user-specific attributes in developing a customized cosmetic recommendation system.*
Keywords: Sentiment analysis, Hybrid recommendation, SVD, KoBERT

1. **Introduction.** With the recent advancement of Artificial Intelligence (AI) technologies, data-driven decision-making has gained increasing importance in the beauty and healthcare industries, particularly for the provision of personalized services [1]. In the cosmetics domain, where individual skin characteristics such as concerns, tone, and sensitivity vary significantly, there is a growing demand for sophisticated recommendation systems that incorporate not only user experience but also emotional satisfaction. Conventional recommendation approaches predominantly rely on simple collaborative filtering or content-based methods grounded in ratings, purchase history, or popularity metrics. However, these approaches fail to sufficiently account for users' skin conditions, affective preferences, and product usage contexts, resulting in limited precision and persuasiveness of recommendations.

This limitation is particularly evident in the case of functional cosmetics designed to address specific concerns such as acne reduction, sensitivity improvement, whitening, or anti-aging. For such products, the quality of recommendations can vary substantially depending on the interaction between complex user conditions and emotional responses. Therefore, rating-based single-dimensional approaches are insufficient for predicting actual satisfaction or product suitability. To overcome these shortcomings, a multidimensional recommendation strategy that integrates text review analysis, skin profile information, and user evaluation patterns is required.

DOI: 10.24507/icicelb.17.08.xxx

This study proposes a hybrid recommendation model that combines semantic analysis of text reviews, sentiment scores, skin type and user profiles, and collaborative filtering-based evaluation patterns. Core nouns were extracted using the Komoran morphological analyzer, and a sentiment lexicon was constructed to analyze review texts. Contextual similarity was computed using KoBERT and incorporated into the content-based filtering process. In parallel, sentiment-adjusted ratings (rating_aug) were applied to training a Singular Value Decomposition (SVD)-based collaborative filtering model, thereby enhancing overall recommendation quality.

For effective text analysis, this study referred to several prior works. Moon et al. proposed a personalized recommendation system based on collaborative filtering [2]. Kim et al. introduced a recommendation approach that utilizes deep learning techniques by incorporating product search history and contextual information such as users' skin types and concerns [3]. Reimers and Gurevych presented an efficient sentence embedding method using Sentence-BERT (SBERT) [4]. Finally, the authors' previous study was referenced to support effective preprocessing and stopword handling [5]. Mishra presented BERTopic, a novel methodology for discovering potential insights by performing topic modeling [6]. Gan et al. evaluated the independence (separability) between topic clusters to gain a better understanding of the structure and content of text data [7].

By integrating sentiment analysis, user skin characteristics, and evaluation patterns within a hybrid architecture, this study aims to address the limitations of traditional recommendation systems and contribute to the realization of personalized recommendations for functional cosmetics. The subsequent sections present a review of related research, model design, experimental procedures, and performance evaluation to discuss the effectiveness and applicability of the proposed model.

2. Research Methodology and Analysis.

This section describes the overall research framework, data preprocessing procedures, and the hybrid modeling approach combining KoBERT and SVD for personalized cosmetic recommendations.

2.1. Sentiment analysis.

Sentiment analysis refers to a set of NLP techniques used to identify, classify, and quantify subjective emotions, attitudes, opinions, and evaluations embedded in textual data. It is primarily used to determine the polarity of sentiment (such as positive, negative, or neutral) or to quantify the intensity of emotional expression. This method is widely applied to unstructured data including online reviews, social media posts, survey responses, and customer feedback to extract meaningful insights [8].

Approaches to sentiment analysis are generally categorized into lexicon-based methods and machine learning or deep learning-based methods. Lexicon-based approaches determine sentiment polarity by matching text against pre-defined sentiment lexicons composed of positive and negative words. This method offers advantages in terms of structural simplicity and interpretability. In contrast, machine learning and deep learning approaches can learn from large volumes of text data, enabling the detection of contextual meaning and complex emotional expressions. These approaches can also account for interactions among words and variations in linguistic nuance.

Sentiment can be analyzed at the word, sentence, paragraph, or document level, and recent advancements have integrated contextual embedding models that allow for the processing of metaphorical expressions, mixed emotions, and polysemous words. The applicability of sentiment analysis has expanded across domains such as recommender systems, marketing strategy, consumer experience analysis, policy evaluation, and healthcare. It is increasingly recognized as a core tool for capturing users' psychological satisfaction and experiential perceptions that cannot be fully observed through quantitative ratings or behavioral choices alone.

2.2. **Singular value decomposition.** Singular Value Decomposition (SVD) is a fundamental linear algebra technique that decomposes a real or complex matrix into three constituent matrices, enabling dimensionality reduction and the extraction of latent structural features [9]. A given $m \times n$ matrix A can be decomposed as follows:

$$A = U\Sigma V^T$$

Here, U is an $m \times m$ orthogonal matrix representing the principal directions in the data, while V is an $n \times n$ orthogonal matrix that captures the axis directions in the variable space. Σ is an $m \times n$ diagonal matrix whose entries are singular values, indicating the contribution of each latent dimension to the overall variance of the data.

SVD is widely used for dimensionality reduction, noise reduction, data compression, and Latent Semantic Analysis (LSA) because it allows high-dimensional matrices to be approximated in lower-dimensional spaces while preserving essential information. In text mining, for instance, applying SVD to a term-document matrix enables the discovery of latent semantic structures, improving the density and interpretability of sparse representations.

Furthermore, SVD generalizes the concept of eigendecomposition, making it applicable even to non-square matrices with high numerical stability. By summarizing dominant patterns in the data, SVD helps mitigate the curse of dimensionality and enhances the performance of downstream tasks such as clustering, classification, and recommender systems.

2.3. **KoBERT.** KoBERT (Korean Bidirectional Encoder Representations from Transformers) is a pretrained language model designed specifically for Korean Natural Language Processing (NLP). Developed by SKT Brain [10], the model is based on the BERT architecture but trained on Korean-language corpora. Unlike the original BERT, which is predominantly pretrained on English text, KoBERT is trained on large-scale Korean datasets sourced from sources such as Wikipedia, news articles, and online communities, thereby enhancing its ability to capture semantic representations at the sentence level.

In terms of architecture, KoBERT employs a bidirectional Transformer-based encoder and utilizes the SentencePiece tokenizer in place of WordPiece. This tokenizer is better suited to the linguistic characteristics of Korean, effectively handling grammatical particles, verb endings, and compound words. As a result, KoBERT can infer dependency relationships between words within a sentence more precisely and demonstrates superior generalization performance compared to models based on morphological unit tokenization.

KoBERT can be applied to a wide range of downstream NLP tasks, including sentiment analysis, sentence classification, Named Entity Recognition (NER), Natural Language Inference (NLI), and semantic similarity assessment. By fine-tuning the pretrained embeddings, the model can achieve high performance even with limited training data. In sentiment analysis in particular, KoBERT enables the generation of sentence vectors that reflect contextual meaning, thereby overcoming the limitations of frequency-based approaches that rely solely on emotion-related keywords.

Due to these characteristics, KoBERT is highly effective for learning semantic representations from unstructured Korean text such as reviews, comments, and consultation records. It also holds strong potential for applications in user experience-based recommender systems, opinion mining, and emotion-aware interaction models.

3. **Research Results.** This study implements a personalized cosmetic recommendation system through a series of processes encompassing web-based review data collection, sentiment analysis, integration of skin-specific attributes, and the development of a hybrid recommendation model. The overall research procedure consists of five stages: 1) data preprocessing, 2) sentiment analysis and data augmentation, 3) skin profile filtering,

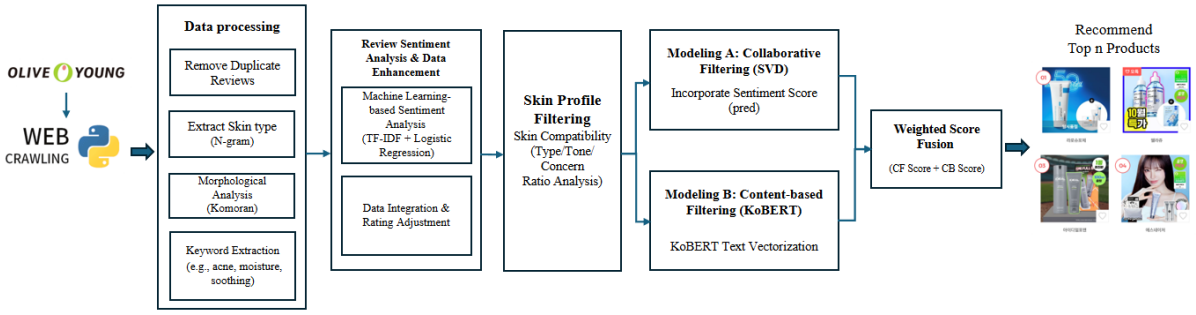


FIGURE 1. Research process

4) integration of recommendation models, and 5) generation of final recommendation outcomes. Figure 1 shows the overall process of this study.

3.1. **Data preprocessing, review sentiment analysis and data augmentation.** Based on review data crawled from the Olive Young website, a series of preprocessing steps was conducted. First, reviews identified as duplicates or written by the same user were removed to ensure data reliability and density. Next, the N-gram technique was employed to extract expressions related to skin type, tone, and concerns (e.g., sensitivity, dryness, and acne) from the review texts.

Morphological analysis was performed using the Komoran analyzer, and a keyword set centered on nouns was constructed to refine the semantic units of the reviews. Subsequently, stopwords were removed, and dictionary-based filtering with core terms (e.g., acne, hydration, and soothing) was applied to building the input structure for the recommendation model. A total of 12,347 review instances were collected, of which 522 were removed through duplicate elimination. Consequently, 11,795 reviews were used in this study. Table 1 presents the attribute columns of the review dataset employed in the analysis. To ensure the reliability of the review data, the authors manually checked the content and positive/negative matching.

TABLE 1. Data columns and descriptions

Name	Detail
item_id	Product data
user_id	User ID
skin_type	Skin type (dry, oily, etc.)
skin_tone	Brightness, darkness, etc.
skin_concerns	Sensitivity, dehydration, acne, etc.
review	User comments
rating	Score ($1 \sim 5$)

Sentiment analysis was conducted on the refined review data to generate sentence-level polarity scores for positive and negative evaluations. In the initial stage, a rule-based approach grounded in a sentiment lexicon was applied. Subsequently, a machine learning classification model combining TF-IDF vectorization and logistic regression was employed to enhance the precision of the sentiment scores. The sentiment outputs were then integrated with the original rating data to perform rating augmentation, and the resulting values were incorporated into the training of the collaborative filtering model.

3.2. **Skin profile filtering and recommendation model configuration.** Explicit user attributes such as skin type, skin tone, and primary skin concerns were first used to perform a pre-filtering step. For each review or product category, compatibility ratios (e.g.,

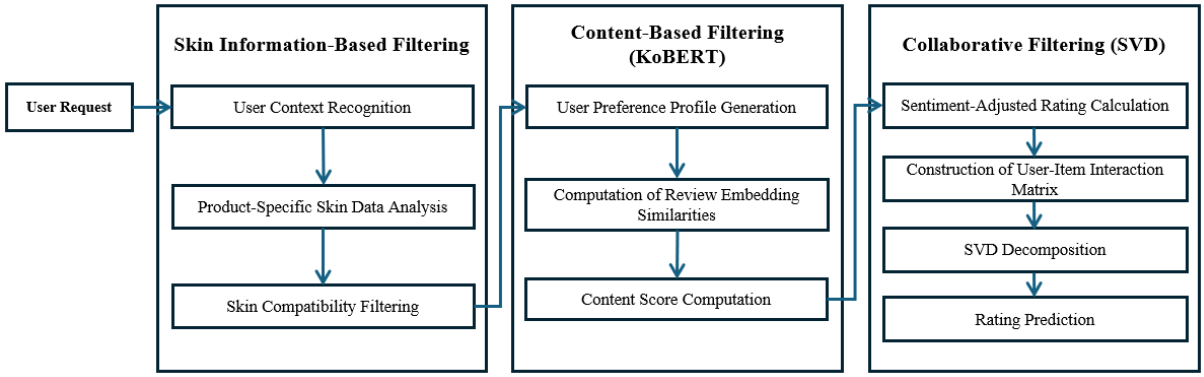


FIGURE 2. Detailed modeling process

dry/oily skin, sensitivity, and acne-prone conditions) were calculated to reduce the exploration space of candidate data. This process not only enhanced the level of personalization in the recommendations but also reduced unnecessary computational overhead. Figure 2 shows the skin suitability filtering and detailed modeling process of this study.

Based on the refined candidate set, two recommendation modules were designed in parallel.

- Model A: Collaborative Filtering via SVD

A user-item rating matrix was factorized using Singular Value Decomposition (SVD) to extract latent factors. The sentiment-adjusted ratings (predicted values) generated in the previous stage were incorporated, enabling recommendations that reflect emotional satisfaction in contrast to traditional rating-based collaborative filtering approaches.

To validate the selection of SVD for the collaborative filtering module (Model A), its performance was benchmarked against other common collaborative filtering algorithms, namely K-Nearest Neighbors (KNN) and Non-negative Matrix Factorization (NMF). Predictive accuracy was evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), where lower values indicate superior performance.

TABLE 2. Collaborative filtering model performance comparison

	SVD	KNN	NMF
RMSE	0.5481	0.5717	0.6594
MAE	0.3095	0.3217	0.4418

As shown in Table 2, SVD achieved the lowest error rates (RMSE = 0.5481, MAE = 0.3095), outperforming both KNN and NMF. This superior predictive accuracy demonstrates SVD's effectiveness in extracting latent factors from the sentiment-adjusted user-item matrix.

- Model B: Content-Based Filtering with KoBERT

KoBERT embeddings were applied to sentiment-bearing sentences and product reviews to generate text vectors that preserve contextual meaning. This enabled similarity-based content recommendations beyond simple keyword matching. Figure 3 presents the proportion of the top 10 product reviews by skin type derived from the modeling process described above.

3.3. **Score weighting combination and final recommendation results.** The results from collaborative filtering (Model A score) and content-based filtering (Model B score) were integrated using a weighted ensemble approach. The final aggregated scores were then ranked based on the top-n items, producing personalized product recommendations

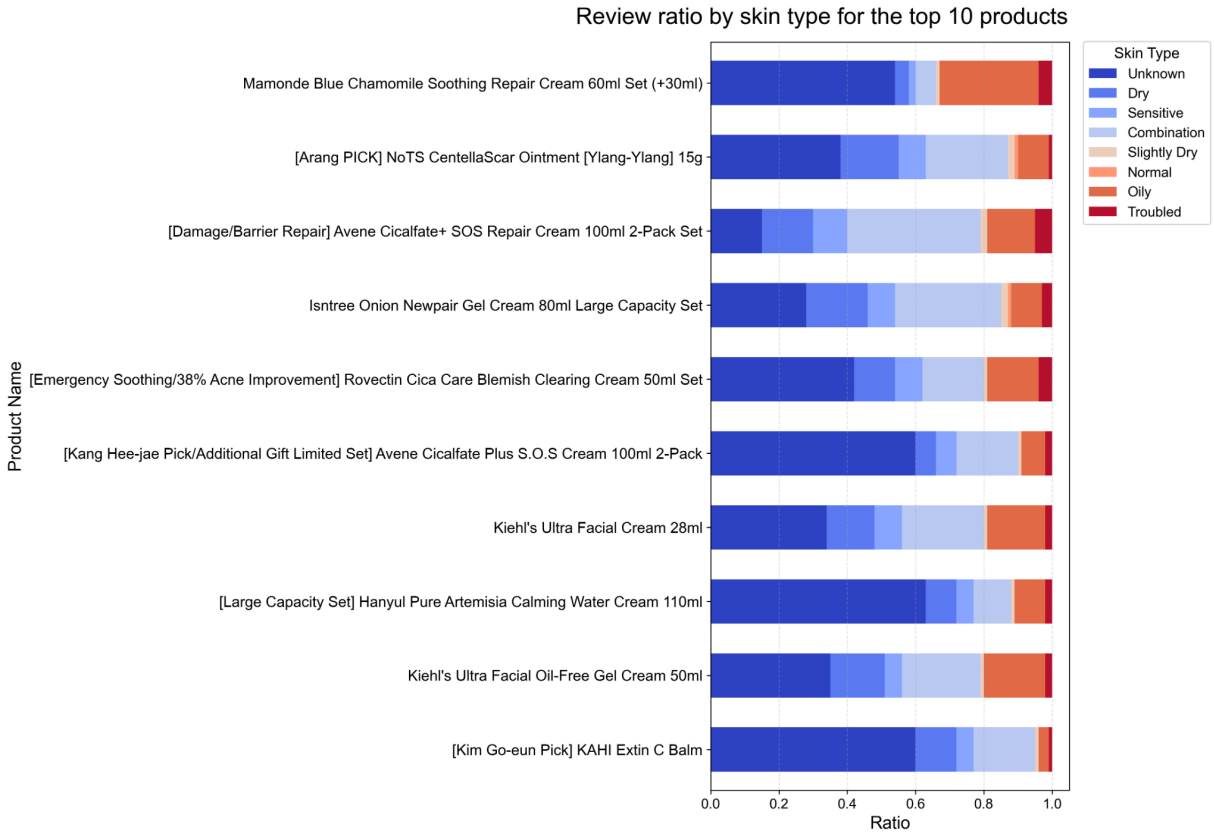


FIGURE 3. (color online) Review ratio by skin type for the top 10 products

	item_id	final_score
0	Mamonde Blue Chamomile Soothing Repair Cream 6...	0.879244
1	[Arang PICK] NoTS CentellaScar Ointment [Ylang...	0.832843
2	[Damage/Barrier Repair] Avene Cicalfate+ SOS R...	0.752454
3	Isntree Onion Newpair Gel Cream 80ml Large Cap...	0.726204
4	[Emergency Soothing/38% Acne Improvement] Rove...	0.654759
5	[Kang Hee-jae Pick/Additional Gift Limited Set...	0.621607
6	Kiehl's Ultra Facial Cream 28ml	0.599900
7	[Large Capacity Set] Hanyul Pure Artemisia Cal...	0.578260
8	Kiehl's Ultra Facial Oil-Free Gel Cream 50ml	0.400000
9	[Kim Go-eun Pick] KAHI Extin C Balm	0.362142

FIGURE 4. Final personalized recommendation results based on the integrated score

that reflect both the user's skin characteristics and their sentiment tendencies in reviews. Figure 4 shows the results of customized product recommendations. The final_score in Figure 4 indicates the accuracy of how suitable the recommended product is for the item_id.

4. **Conclusion.** This study proposed a personalized cosmetic recommendation system that integrates user-generated review data, sentiment analysis, skin profile filtering, and a hybrid recommendation model. By combining enhanced rating information derived from

sentiment scores with both collaborative filtering and content-based filtering, the system was designed to capture not only explicit user preferences but also implicit emotional responses embedded in textual feedback.

The preprocessing pipeline including duplicate removal, morphological analysis, and keyword-based filtering ensured the reliability and semantic clarity of the review dataset. Sentiment analysis using a rule-based approach and a machine learning classifier further refined rating accuracy, enabling sentiment-adjusted predictions within the SVD-based collaborative filtering model. In parallel, KoBERT embeddings supported context-aware content-based recommendations by preserving linguistic nuances in user reviews.

A skin profile filtering mechanism reduced the search space by aligning candidate products with individual skin characteristics, thereby increasing personalization and computational efficiency. The final ensemble model weighted the outputs of the collaborative filtering and content-based modules to generate customized recommendations sorted by an integrated score.

Future work may extend this approach with multimodal data sources, such as ingredient compositions, dermatological test results, or real-time user feedback, to enhance accuracy and generalizability. Additionally, expanding the evaluation with user studies or A/B testing could provide further validation of the system's practical applicability.

Acknowledgment. This research was supported by the Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (P0017123, The Competency Development Program for Industry Specialist) and by the G-LAMP Program of the National Research Foundation of Korea (NRF) grant funded by the Ministry of Education (No. RS-2025-25441317).

REFERENCES

[1] M. Lee, Artificial intelligence (AI) emerging as the next-generation beauty trend, *Patent News*, vol.7, no.155, pp.82-83, 2018.

[2] H. Moon, J. Lim, D. Kim and Y. Cho, A deep learning based recommender system using visual information, *Knowledge Management Research*, vol.21, no.3, pp.27-44, 2022.

[3] H. Kim, W. Shin, D. Shin and H.-W. Kim, Beauty product recommendation system using customer attributes information, *Information Systems Review*, vol.23, no.4, pp.69-84, 2021.

[4] N. Reimers and I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-Networks, *arXiv Preprint*, arXiv: 1908.10084, 2019.

[5] H. Kim, C. Jeon and D. Kim, A study on the satisfaction of delivery app users by analyzing app market review data: Focusing on BaeMin, Coupang Eats, *The Journal of Society for e-Business Studies*, vol.27, no.3, pp.113-125, 2022.

[6] M. Mishra, A holistic review of customer experience research: Topic modelling using BERTopic, *Marketing Intelligence & Planning*, vol.43, no.4, pp.802-820, 2025.

[7] L. Gan, T. Yang, Y. Huang, B. Yang, Y. Luo, L. Richard and D. Guo, Experimental comparison of three topic modeling methods with LDA, Top2Vec and BERTopic, *International Symposium on Artificial Intelligence and Robotics*, pp.376-391, 2023.

[8] Y. Mejova, *Sentiment Analysis: An Overview*, Survey Paper, Department of Computer Science, University of Iowa, 2009.

[9] A. Akritas and G. Malaschonok, Applications of singular-value decomposition (SVD), *Mathematics and Computers in Simulation*, vol.67, nos.1-2, pp.15-31, 2004.

[10] SKT Brain, Korean BERT pre-trained cased (KoBERT), *GitHub*, https://github.com/SKTBrain/KoBERT, 2019.